Green University of Bangladesh
Department of Computer Science and Engineering (CSE)

Faculty of Sciences and Engineering
Semester: (Spring, Year:2026), B.Sc. in CSE

Lab Report NO 01
Course Title: Database System Lab
Course Code: CSE 210 Section: 242D3

Lab Experiment Name: Creating and Managing a Library Database with Constraints and Table Alterations in SQL

Student Details

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Lab Date : 26/02/2026
Submission Date : 05/03/2026
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Lab Report Status

Marks: …… **Signature:.....**
Comments:..... **Date:.....**

1. TITLE OF THE LAB EXPERIMENT

Creating and Managing a Library Database with Constraints and Table Alterations in SQL

2. OBJECTIVES

- To create a database named library_lab.
- To create tables MEMBER, BOOK, and LOAN with appropriate constraints.
- To apply Primary Key, Foreign Key, UNIQUE, and CHECK constraints.
- To insert records into the tables and observe constraint validation.
- To generate errors for invalid data such as duplicate emails, invalid foreign keys, and negative prices.
- To modify table structures using ALTER TABLE commands.
- To rename tables and modify existing columns.

3. PROCEDURE / ANALYSIS / DESIGN

I. Open the SQL environment (MySQL / SQL Server / Oracle depending on lab).
II. Create a database named library_lab.
III. Use the created database.
IV. Create three tables: MEMBER, BOOK, and LOAN.
V. Apply constraints:
    - Primary Key for unique identification.
    - Foreign Key to maintain relationships.
    - UNIQUE for fields like Email and ISBN.
    - CHECK constraints for Age and Price validation.
VI. Insert sample data into the tables.
VII. Attempt invalid operations to observe constraint errors.
VIII. Modify the structure of the tables using ALTER TABLE commands.
IX. Rename the table LOAN to BORROW.
X. Verify all changes using DESCRIBE or SELECT statements.

4. IMPLEMENTATION

The implementation was performed using SQL. Below are the structural details for the most critical parts:

Code:

```
CREATE DATABASE library_lab;
Use library_lab;
```

Member Table :

```
CREATE TABLE MEMBER(
   MemberID int AUTO_INCREMENT,
   FullName varchar(50) NOT NULL,
   Email varchar(50) NOT NULL,
   Age int,
   Gender ENUM ('Male', 'Female', 'Other'),
   PRIMARY KEY (MemberID),
   UNIQUE(Email),
   CHECK(Age>=12)
   )


INSERT INTO MEMBER (FullName, Email, Age, Gender)
VALUES('Nusrat', 'nusrat@gmail.com', 20, 'Female'),
('Saif', 'saif@gmail.com', 22, 'Male'),
('Sakib', 'sakib@gmail.com', 15, 'Male'),
('Sumaiya', 'sumaiya@gmail.com', 21, 'Female'),
('Noyon', 'noyon@gmail.com', 23, 'Male')
```

Book Table :

```
CREATE TABLE BOOK (
   BookID INT AUTO_INCREMENT,
   Title VARCHAR(255) NOT NULL,
   ISBN VARCHAR(20) NOT NULL,
   Price int,
   Book_Status ENUM('AVAILABLE', 'BORROWED') DEFAULT 'AVAILABLE',
   PRIMARY KEY (BookID),
   UNIQUE (ISBN)
   )


INSERT INTO BOOK (Title, ISBN, Price)
VALUES ('The Great Gatsby', '9780743273565', 159),
('1984', '9780451524935', 125),
('The Hobbit', '9780547928227', 200),
('Sapiens', '9780062316097', 180),
('SQL Guide', '9781234567890', 450)
```

Loan Table :

```
CREATE TABLE LOAN (
   LoanID INT AUTO_INCREMENT,
   MemberID INT,
   BookID INT,
   PRIMARY KEY (LoanID),
   FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID),
   FOREIGN KEY (BookID) REFERENCES BOOK(BookID)
)


INSERT INTO LOAN (MemberID, BookID)
VALUES (1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5),
(1, 2);
```

Add Phone column in member Table :

```
ALTER TABLE MEMBER ADD Phone VARCHAR(15) NOT NULL DEFAULT 'N/A';

UPDATE MEMBER SET PHONE = '01471245451' WHERE FullName = 'Nusrat';
UPDATE MEMBER SET PHONE = '01712545455' WHERE FullName = 'Saif';
UPDATE MEMBER SET PHONE = '01922415454' WHERE FullName = 'Sakib';
UPDATE MEMBER SET PHONE = '01385124457' WHERE FullName = 'Sumaiya';
UPDATE MEMBER SET PHONE = '01812545455' WHERE FullName = 'Noyon';
```

Modify Price in book Table :

```
ALTER TABLE BOOK MODIFY Price DECIMAL(8, 2);
```

Change column name in book Table :

```
ALTER TABLE BOOK CHANGE Status Availability ENUM('AVAILABLE',
'BORROWED') DEFAULT 'AVAILABLE';
```

Drop Gender column in member Table :

```
ALTER TABLE MEMBER DROP COLUMN Gender;
```

Rename loan Table :

```
RENAME TABLE loan to BORROW;
```

5. TEST RESULT / OUTPUT

The following snapshots represent the "Browsed" state of the database after successful execution.

	Table	Action	Rows	Type	Collation	Size	Overhead
☐	book	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	loan	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	member	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
	3 tables	Sum	16	InnoDB	utf8mb4_general_ci	112.0 KiB	0 B

Check all With selected:

Snapshot 1: library_lab Database with Table

			MemberID	FullName	Email	Age	Gender
☐ Edit	Copy	Delete	1	Nusrat	nusrat@gmail.com	20	Female
☐ Edit	Copy	Delete	2	Saif	saif@gmail.com	22	Male
☐ Edit	Copy	Delete	3	Sakib	sakib@gmail.com	15	Male
☐ Edit	Copy	Delete	4	Sumaiya	sumaiya@gmail.com	21	Female
☐ Edit	Copy	Delete	5	Noyon	noyon@gmail.com	23	Male

Check all *With selected:* Edit Copy Delete Export

Snapshot 2: Book Table

	BookID	Title	ISBN	Price	Book_Status
Edit Copy Delete	1	The Great Gatsby	9780743273565	159	AVAILABLE
Edit Copy Delete	2	1984	9780451524935	125	AVAILABLE
Edit Copy Delete	3	The Hobbit	9780547928227	200	AVAILABLE
Edit Copy Delete	4	Sapiens	9780062316097	180	AVAILABLE
Edit Copy Delete	5	SQL Guide	9781234567890	450	AVAILABLE

Check all *With selected:* Edit Copy Delete Export

Snapshot 3: Loan Table

	LoanID	MemberID	BookID
Edit Copy Delete	1	1	1
Edit Copy Delete	2	2	2
Edit Copy Delete	3	3	3
Edit Copy Delete	4	4	4
Edit Copy Delete	5	5	5
Edit Copy Delete	6	1	2

Check all *With selected:* Edit Copy Delete Export

Snapshot 4: Add Phone column in member Table

	MemberID	FullName	Email	Age	Gender	Phone
Edit Copy Delete	1	Nusrat	nusrat@gmail.com	20	Female	01471245451
Edit Copy Delete	2	Saif	saif@gmail.com	22	Male	01712545455
Edit Copy Delete	3	Sakib	sakib@gmail.com	15	Male	01922415454
Edit Copy Delete	4	Sumaiya	sumaiya@gmail.com	21	Female	01385124457
Edit Copy Delete	5	Noyon	noyon@gmail.com	23	Male	01812545455

Check all *With selected:* Edit Copy Delete Export

Snapshot 5: Error for email duplicate

Error

SQL query: Copy

```
INSERT INTO MEMBER (FullName, Email, Age)
VALUES('Farin', 'nusrat@gmail.com', 20);
```

MySQL said:

```
#1062 - Duplicate entry 'nusrat@gmail.com' for key 'Email'
```

Snapshot 6: Change column name in book Table

			Team_ID	Team_Name	Player_ID	Player_Name
Edit	Copy	Delete	1	Tigers	101	Shakib
Edit	Copy	Delete	2	Lions	102	Saif
Edit	Copy	Delete	3	Eagles	103	Shourov
Edit	Copy	Delete	4	Warriors	104	Mofid
Edit	Copy	Delete	5	Panthers	105	Tamim

Check all *With selected:* Edit Copy Delete Export

Snapshot 7: Drop Gender column in member Table

←T→	MemberID	FullName	Email	Age	Phone
Edit Copy Delete	1	Nusrat	nusrat@gmail.com	20	01471245451
Edit Copy Delete	2	Saif	saif@gmail.com	22	01712545455
Edit Copy Delete	3	Sakib	sakib@gmail.com	15	01922415454
Edit Copy Delete	4	Sumaiya	sumaiya@gmail.com	21	01385124457
Edit Copy Delete	5	Noyon	noyon@gmail.com	23	01812545455

Check all *With selected:* Edit Copy Delete Export

Snapshot 8: Modify Price in book Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	BookID	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	Title	varchar(255)	utf8mb4_general_ci		No	*None*			Change Drop More
3	ISBN	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
4	Price	decimal(8,2)			Yes	*NULL*			Change Drop More
5	Book_Status	enum('AVAILABLE', 'BORROWED')	utf8mb4_general_ci		Yes	AVAILABLE			Change Drop More

Snapshot 9: Rename loan Table

Table	Action	Rows	Type	Collation	Size	Overhead
book	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
borrow	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	48.0 KiB	-
member	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
3 tables	Sum	16	InnoDB	utf8mb4_general_ci	112.0 KiB	0 B

Check all With selected:

6. ANALYSIS AND DISCUSSION

This experiment demonstrates how relational databases enforce data integrity through constraints. Primary keys ensure each record is unique, while foreign keys maintain relationships between tables. UNIQUE constraints prevent duplicate entries, and CHECK constraints validate logical conditions such as age or price limits.

Error testing is important because it confirms that constraints are working properly. When invalid data is inserted, the database system rejects it, ensuring reliable data storage.
Using ALTER TABLE allows modification of database structures without deleting existing data. This flexibility is essential when database requirements change over time.

7. SUMMARY

In this lab experiment, a library management database was designed and implemented using SQL. Tables for members, books, and loans were created with various constraints to maintain data integrity. Sample records were inserted and errors were tested to verify constraint functionality. Finally, several ALTER TABLE operations were performed to modify the database structure and rename a table. This experiment highlights the importance of constraints and schema modification in relational database management.